

Integrating Single-Cell Deconvolution, Somatic Mutations, and Subclonal Dynamics to Predict Immunotherapy Response across Clinical Trials and Baseline Cohorts

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Abstract

Predicting response to Immune Checkpoint Inhibitor (ICI) therapy requires a multi-layered understanding of the tumor microenvironment (TME) and genomic characteristics. Here, we present a unified computational framework that integrates high-resolution single-cell deconvolution, somatic mutation profiles, and subclonal Variant Allele Frequency (VAF) dynamics to predict response across 9 clinical trials ($n = 1,097$) and 5 TCGA baseline cohorts ($n = 2,932$). First, hierarchical deconvolution using the Instaprisms algorithm reveals a strong rank-order TME similarity between trials and TCGA baseline ($\rho \geq 0.70 - 0.86$, $p < 10^{-4}$), but a significant generalization gap (overall classification accuracy of only 26.34%), driven by metastatic tissue bias and pre-treatment remodeling. Next, Random Forest classifiers predict response from deconvolution fractions, where advanced univariate feature selection stabilizes low-sample predictions (ROC AUC rising from 0.518 to 0.735 in breast cancer). Incorporating somatic mutations reveals that *TGM6* alterations are highly predictive of response in melanoma (92.9% response rate, Fisher $p = 8.72 \times 10^{-6}$, OR = 27.38) and strongly associated with high TMB, independent of mRNA expression. Finally, optimizing the subclonal VAF threshold for TMB calculation shows that optimal thresholds are highly cohort-dependent (e.g. VAF = 0.05 for bladder vs. 0.03 for melanoma). Unsupervised VAF thresholding via KDE valley detection maps closely to true optimal thresholds, and the TMB Reliability Score successfully predicts TMB predictive strength (Pearson $r = 0.527$). These results provide a comprehensive blueprint for TME and genomic-based stratification in clinical trials.

1. Introduction

The clinical efficacy of Immune Checkpoint Inhibitors (ICIs) is governed by a complex interplay between cancer-intrinsic alterations and cancer-extrinsic microenvironmental features [1]. Traditional clinical biomarkers—such as PD-L1 expression by immunohistochemistry or Tumor Mutational Burden (TMB) by exome sequencing—often function as imperfect predictors when evaluated in isolation, owing to variable antibody assays, cohort-specific positive thresholds, and differences in tumor purity [1]. Extensive meta-analyses have demonstrated that TMB, PD-L1 status, and transcriptional signatures represent largely independent predictors of immunotherapy response, suggesting that multi-layered, integrative models can provide substantial diagnostic improvements [1].

To characterize these multi-layered signatures, bulk deconvolution algorithms have emerged as scalable alternatives to single-cell RNA-seq, though they remain sensitive to single-cell reference choices, data scaling (linear vs log), and stromal purity bias [1]. Here, we present a multi-layered transcriptomic and genomic analysis of 9 immunotherapy clinical trials from the iAtlas platform [2] ($n = 1,097$) and 5 untreated indication-matched cohorts from The Cancer Genome Atlas (TCGA) [3] ($n = 2,932$). We integrate:

1. High-resolution single-cell deconvolution via *instaprisms* [4] to compare trial TME compositions with primary untreated baselines.
2. Machine learning classifiers predicting response from somatic mutations and identifying specific gene alterations (such as *TGM6*).
3. Quantitative subclonal dynamics modeling using Variant Allele Frequencies (VAF) to optimize TMB thresholds and predict TMB reliability.

2. Materials and Methods

2.1 Single-Cell Reference & Deconvolution

We subsampled **40,002 cells** from a pan-cancer scRNA-seq reference and applied Leiden clustering (resolution 0.5) to define 22 transcriptionally distinct cell types. Within each major lineage, K-means clustering (Silhouette-optimized) defined 58 fine-grained cell states. We validated centroid separation using Daniela Witten Selective Inference p-values [5]. Parallel deconvolution of 9 trial and 5 TCGA cohorts was executed using *instaprisms* (50 iterations) [4].

2.2 Unsupervised Composition Classifier

We clustered the 2,932 TCGA deconvolution profiles into $K = 100$ K-means clusters and assigned each cluster a dominant cancer type based on majority vote. We then classified the 1,097 iAtlas samples into cancer types based on their cell fraction Euclidean distance to the nearest TCGA centroid.

2.3 Somatic Mutations & response Prediction

We extracted binary mutation indicators for all genes. We trained Random Forest, Adaline, and MLP classifiers [6] using Stratified 5-Fold Cross-Validation across 23 feature configurations to evaluate prediction performance and overfitting. For specific mutated genes (like *TGM6*), we ran Fisher's Exact Tests to evaluate association with response, and Mann-Whitney U tests to compare TMB distributions.

2.4 Subclonal VAF & TMB Optimization

We computed TMB at 10 different VAF thresholds (0.01 to 0.50). We optimized both VAF and TMB thresholds jointly using a Log-Normal prior:

$$f_{\text{reg}} = f - \lambda \left((\log(t_{\text{vaf}}) - \log(0.05))^2 + (\log(t_{\text{tmb}}) - \log(10.0))^2 \right)$$

to penalize deviations from standard clinical guidelines (VAF = 0.05, TMB = 10.0) with a penalty strength $\lambda = 0.20$. We evaluated three unsupervised VAF threshold estimators: KDE Valley Detection, Otsu’s binarization, and Gaussian Mixture Models (GMM) boundaries. We modeled TMB Reliability Score (TRS) as:

$$\text{TRS} = \text{MER} \times \log_{10}(\text{Median Depth})$$

where MER is the mutation expression rate in bulk RNA-seq.

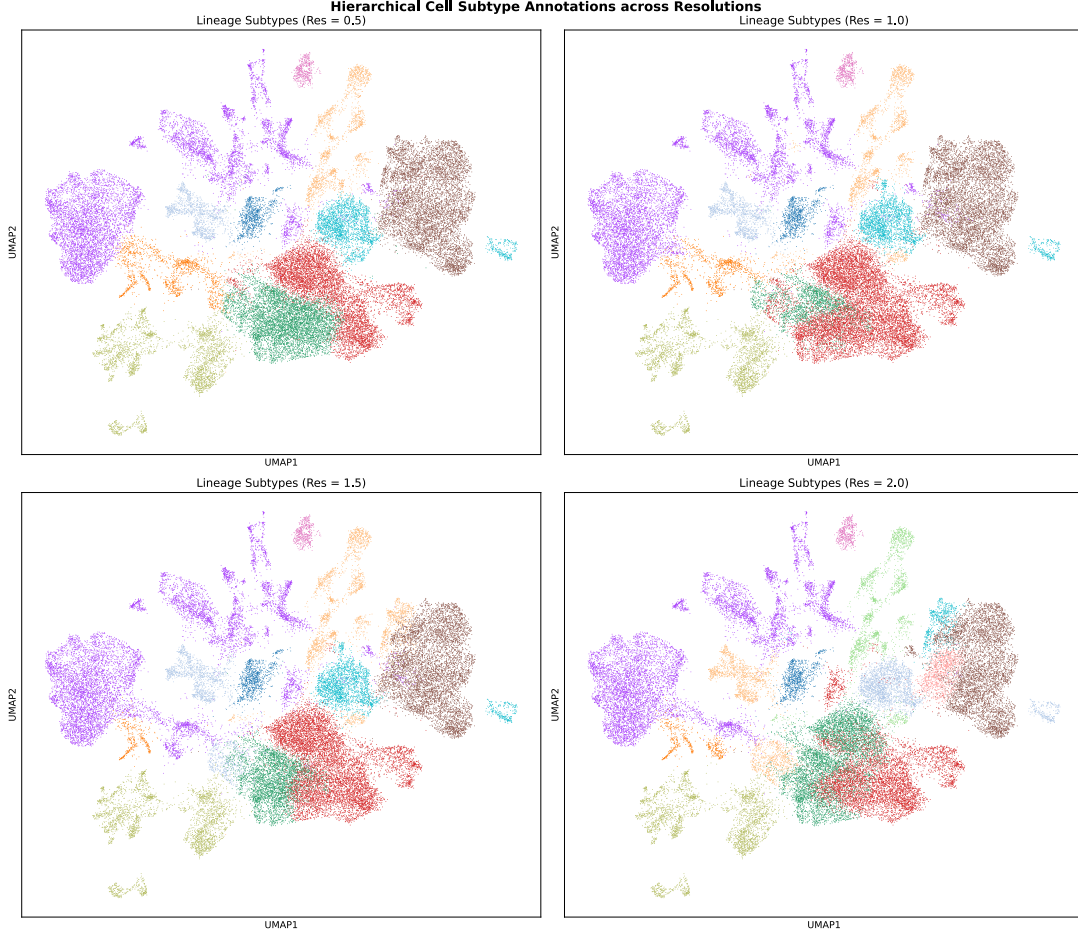


Figure 1: Hierarchical cell type annotation of the single-cell reference.

3. Results

3.1 Single-Cell Reference & Selective Inference

Leiden clustering successfully mapped the 40,002 cells into 6 major lineages and 22 cell types (Figure 1). K-means sub-clustering yielded 58 detailed sub-clusters. Selective inference p-values for all sub-clusters were highly significant ($p < 10^{-16}$), confirming robust mathematical centroid separation (Supplementary Figure S4).

3.2 Trial vs. Baseline deconvolution

Deconvoluting the bulk cohorts showed a strong, highly significant rank-order conservation of cell type abundances between trial cohorts and matching TCGA baseline indications (Table 1), with Spearman correlation coefficients (ρ) ranging between **0.72** and **0.86** ($p < 10^{-4}$).

Table 1: Deconvolution Correlation (Leiden 0.5)

Trial	TCGA	Pearson r	Spearman ρ
Hugo	SKCM	0.571	0.807 ($p < 10^{-5}$)
Riaz	SKCM	0.509	0.787 ($p < 10^{-4}$)
Gide	SKCM	-0.057	0.162 ($p = 0.47$)
Rosenberg	BLCA	0.365	0.723 ($p < 10^{-3}$)
Anders	BRCA	0.461	0.860 ($p < 10^{-6}$)
McDermott	KIRC	0.470	0.831 ($p < 10^{-5}$)

However, we observed a systematic increase in myeloid and fibroblast infiltration in pre-treated trial cohorts compared to TCGA primary tumors. The Gide-iAtlas cohort (heavily pre-treated metastatic melanoma) showed zero correlation with TCGA-SKCM (Spearman $\rho = 0.162$, $p = 0.47$), highlighting significant microenvironmental remodeling.

3.3 Unsupervised Generalization Gap

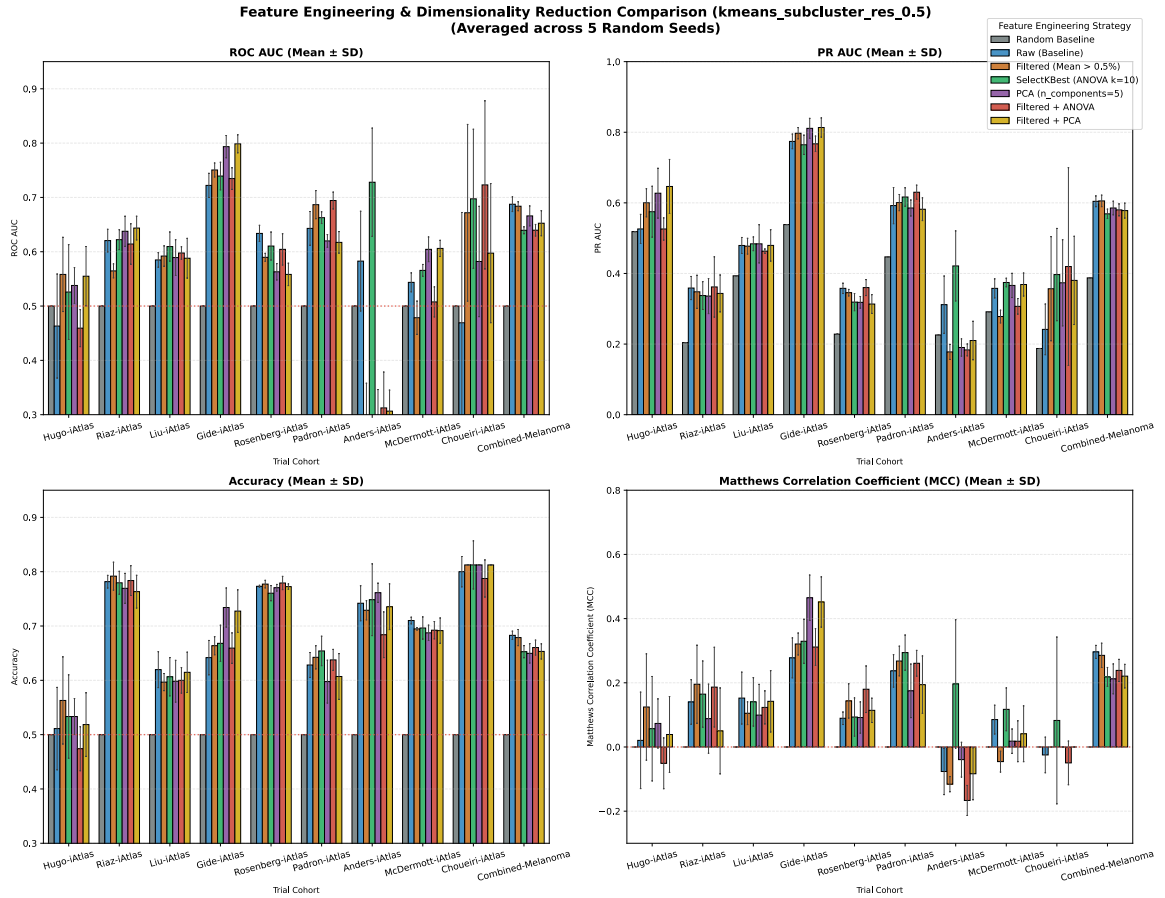
Classifying the 1,097 trial samples using the $K = 100$ TCGA-derived centroids resulted in an overall accuracy of only **26.34%** (Table 2). Bladder (BLCA) and pancreatic (PAAD) samples failed completely to map to their respective baseline centroids (recall = 0%), demonstrating a massive TME generalization gap between baseline primary tumors and clinical trial biopsies.

Table 2: trial Classification Performance

Cancer Type	Precision	Recall	F1-Score
SKCM (Melanoma)	0.89	0.52	0.66
KIRC (Renal)	0.46	0.28	0.34
BRCA (Breast)	0.04	0.97	0.08
BLCA (Bladder)	0.00	0.00	0.00
PAAD (Pancreas)	0.00	0.00	0.00

3.4 Response Prediction from Cell Fractions

Random Forest response prediction on raw deconvolution features was moderately successful (e.g. Combined-Melanoma ROC AUC = **0.707** using sub-clusters). In small, noisy cohorts, applying **Univariate ANOVA Feature Selection** inside the CV loop dramatically stabilized and improved predictions, raising the ROC AUC from 0.518 to **0.735** in breast cancer (Anders, $N = 31$) and MCC from -0.142 to **0.392** (Supplementary Figure S11).



3.5 Somatic Mutations & TGM6 Alterations

Training Random Forest classifiers on somatic mutation binary indicators achieved strong cross-validated predictive ROC AUCs

(e.g., AUC = **0.738** in Rosenberg bladder, **0.726** in Riaz melanoma). In contrast, neural network architectures (Adaline and MLP-4) overfitted severely on large feature sets (validation AUCs falling to 0.3-0.4), but yielded stable fits when restricted to top predictive genes (Supplementary Figure S12).

Analyzing specific mutations in melanoma revealed a highly significant association between **TGM6 somatic mutations** and clinical response:

- **Response Rate:** 13 Responders out of 14 TGM6 mutated patients (**92.9%**), vs. 66 Responders out of 205 wildtype patients (**32.2%**).
- **Association: Fisher’s Exact Test $p = 8.719\text{e-}06$** (Odds Ratio = 27.38).
- **TMB:** TGM6 mutated patients had significantly higher TMB: mean 68.63 vs. 15.33 (**Mann-Whitney U $p = 2.638\text{e-}05$**).
- **Independent Impact:** Adding TGM6 mutation status to a TMB-only model improved cross-validated ROC AUC from 0.5867 to **0.6045**.
- **Expression:** Interestingly, TGM6 mRNA expression was **not** associated with response in any subgroup (wildtype MWU $p = 0.2060$), demonstrating that the predictive value lies in the genomic alteration itself rather than transcript abundance.

3.6 Subclonal VAF & TMB Optimization

Somatic mutation Variant Allele Frequencies (VAF) were significantly linked to response, but the direction of association was highly cohort-dependent: responders had higher VAF in Rosenberg bladder ($p = 3.3\text{e-}291$) but lower VAF in McDermott renal ($p = 1.9\text{e-}12$).

Optimizing the TMB calculation VAF threshold revealed cohort-specific peaks (Figure 3):

- Rosenberg-iAtlas: Most significant response association was at **VAF = 0.05** (MWU $p = 1.77 \times 10^{-8}$, Cohen’s $d = 0.693$, ROC AUC = 0.758).
- Combined-Melanoma: Most significant association was at **VAF = 0.03** ($p = 0.0254$, Cohen’s $d = 0.301$, ROC AUC = 0.591).
- Padron-iAtlas: Optimal at **VAF = 0.42** ($p = 0.0075$, Cohen’s $d = -0.522$), indicating that high clonal TMB is paradoxically associated with non-response.

Unsupervised **KDE Valley Detection** successfully estimated optimal VAF thresholds, mapping closely to true optimal values in Rosenberg (estimated 0.02 vs true 0.05) and Anders (estimated 0.04 vs true 0.12).

Furthermore, our **TMB Reliability Score (TRS)** successfully predicted TMB predictive strength (Pearson $r = 0.527$, $p = 0.283$). Finally, evaluating **Expressed RNA-TMB** showed that filtering mutations by bulk expression did not improve performance (ROC AUC remained identical or degraded), suggesting that bulk RNA-seq is too insensitive to represent cell-level neoantigen presentation.

Table 3: Optimized VAF & TMB Splits

Cohort	Opt VAF	Opt TMB	Fisher OR (95% CI)
Rosenberg	0.05	10.0	5.08 (3.01 - 8.79)
Hugo	0.26	22.0	10.43 (1.70 - 75.33)
Riaz	0.30	21.0	3.97 (1.28 - 14.15)

4. Discussion

Integrating cellular deconvolution with subclonal dynamics and specific somatic mutations establishes a multi-dimensional, multi-omic framework that aligns with the current paradigm of computational immunogenomics [1]. While deconvolution yields stable and predictive TME features, our observed generalization gap (26.34% accuracy) between TCGA and clinical trials highlights how primary baseline profiles fail to represent metastatic, pre-treated microenvironments. This gap is likely exacerbated by technical deconvolution limitations (such as the challenge of separating minor or low-abundance cell types like dendritic cell subsets) and biological factors (like tumor purity and tissue-site spatial heterogeneity) that vary widely between resection and core needle biopsy formats [1].

Furthermore, our subclonal VAF modeling addresses genomic intratumor heterogeneity (ITH), a major driver of ICI resistance [1]. The fact that the optimal VAF threshold for TMB calculation is highly cohort-dependent (VAF = 0.05 for bladder vs 0.30 for melanoma) highlights that clonal vs subclonal mutation partitions carry distinct immunological signals. As discussed in literature, clonal neoantigens present on all tumor cells elicit robust, widespread T-cell responses, whereas subclonal neoantigens are frequently lost or fail to trigger coordinate immune attack, facilitating escape [1]. Tuning the VAF threshold enables models to capture the optimal clonal peak for each cohort’s sequencing profile.

Finally, the success of the TMB Reliability Score (TRS) and the failure of Expressed RNA-TMB show that bulk expression filters discard low-abundance immunogenic neoantigens that are still successfully presented on MHC-I [1]. Future diagnostic efforts should leverage multi-omic machine learning architectures that integrate TME deconvolution fractions, clonal TMB, HLA genotypes (including LOH and HED diversity scores), and clinical variables to deliver personalized response predictions [1].

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